# Object Oriented Programming (CSEG1044)

## Unit 04 – Lecture 03: Lambdas + Functional Interfaces

April 1, 2026

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

# Quick Links

Overview Key Topics Demo Summary

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# Agenda

1 Overview

2 Key Topics

3 Demo / Example

4 Summary

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# Lecture Snapshot

- **Unit focus:** Generics, Lambdas, Swing, and JDBC

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# Lecture Snapshot

- **Unit focus:** Generics, Lambdas, Swing, and JDBC
- **Course thread:** Use Lambdas + Functional Interfaces to move Campus Resource Manager toward a real desktop application with UI, callbacks, and persistence.

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# Recall Before You Start

- **Interfaces**: A lambda only works when a target functional interface defines exactly one abstract method. Main reference: `Unit 02 / Lecture 05 slides/notes`.

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# Recall Before You Start

- **Interfaces**: A lambda only works when a target functional interface defines exactly one abstract method. Main reference: `Unit 02 / Lecture 05 slides/notes`.
- **Anonymous classes**: Lambdas are easier to appreciate once a more verbose callback style is already known. Main reference: `Unit 03 / Lecture 01 slides/notes`.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

# Learning Outcomes

- Identify functional interfaces and explain target typing.

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- Identify functional interfaces and explain target typing.
- Write lambdas for common patterns (comparators, callbacks, runnable tasks).

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# Learning Outcomes

- Identify functional interfaces and explain target typing.
- Write lambdas for common patterns (comparators, callbacks, runnable tasks).
- Apply basic readability rules for when to use lambdas.

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# Key Terms to Know

- **Functional interface**: interface with 1 abstract method (SAM).

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- **SAM**: Single Abstract Method.
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- **Effectively final**: local variable not modified after initialization (can be captured in lambda).

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# Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.

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1. Refresh the prerequisite bridge before reading new ideas in isolation.
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3. Trace the worked example and connect each line back to one concept frame.
4. Attempt the mini exercises before moving to the recap and exit check.

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# Unit 04: Generics, Lambdas, GUI Swing & Database Connectivity

- Lecture focus: Lambdas + Functional Interfaces

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# Today: Roadmap

- Functional interfaces concept; target typing

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# Today: Roadmap

- Functional interfaces concept; target typing
- Lambda syntax and common patterns (comparators, callbacks)

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# Today: Roadmap

- Functional interfaces concept; target typing
- Lambda syntax and common patterns (comparators, callbacks)
- Readability guidelines: when to use / when to avoid

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# Functional interface (SAM)

- Interface with exactly one abstract method.

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- Examples: `Runnable`, `Comparator<T>`, many Swing listeners.

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# Functional interface (SAM)

- Interface with exactly one abstract method.
- Examples: `Runnable`, `Comparator<T>`, many Swing listeners.
- Optional: annotate with `@FunctionalInterface`.

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# Target typing (why lambdas work)

- Lambda gets its type from context (assignment or method parameter).

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# Target typing (why lambdas work)

- Lambda gets its type from context (assignment or method parameter).
- Example: `Runnable r = () -> ...`
- Example: `list.sort((a,b) -> ...)`

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# Anonymous Class → Lambda

- Start from a one-method anonymous class that already matches a functional interface.

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- Keep the parameter list and method body; drop the class boilerplate.

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# Anonymous Class → Lambda

- Start from a one-method anonymous class that already matches a functional interface.
- Keep the parameter list and method body; drop the class boilerplate.
- Result: shorter code, same contract, easier to read when the behavior is small.

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# Lambda syntax (common forms)

- `() -> expr`

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- `() -> expr`
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- `() -> expr`
- `x -> expr`
- `(a, b) -> expr`
- `x -> { ...; return ...; }`

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# Readability guidelines

- Use lambdas for short, clear behavior (1–3 lines).

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- Avoid long lambdas; extract to a named method.
- Use meaningful parameter names if not obvious.

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# Checkpoint and Reflection

## Reflection prompt

Where would *Lambdas + Functional Interfaces* matter most in Campus Resource Manager, and which design choice would you justify using this lecture's ideas?

## Quick self-check

Which part needs one more example before you move on: *Functional interfaces (concept)*, *Target typing (how lambdas get a type)*, the worked example, or the recap?

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# Example: sort with lambda (Comparator)

```
list.sort((a, b) -> Double.compare(b.getCgpa(), a.getCgpa
    ()));
```

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# Example: runnable task

```
Runnable task = () -> System.out.println("Hello from
    thread");
new Thread(task).start();
```

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# Mini Exercises (2–3 mins)

1. What is a functional interface?

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# Mini Exercises (2–3 mins)

1. What is a functional interface?
2. Where does a lambda get its type from?

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# Mini Exercises (2–3 mins)

1. What is a functional interface?
2. Where does a lambda get its type from?
3. When should you avoid lambdas?

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# Watch-outs

- Assuming any interface can be used as a lambda target.

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# Watch-outs

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- Packing too much business logic into one lambda.

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# Watch-outs

- Assuming any interface can be used as a lambda target.
- Packing too much business logic into one lambda.
- Ignoring the target type and getting confused about the method being implemented.

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# Key Takeaways

- Lambdas are short behavior implementations for functional interfaces.

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- Lambdas are short behavior implementations for functional interfaces.
- Target typing infers lambda type from context.
- Keep lambdas short; extract complex logic to methods.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

# Checkpoint Questions

1. Give one example of a functional interface in Java.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

# Checkpoint Questions

1. Give one example of a functional interface in Java.
2. Write a simple lambda for `Runnable`.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

# Checkpoint Questions

1. Give one example of a functional interface in Java.
2. Write a simple lambda for `Runnable`.
3. State one readability rule for lambdas.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

# Next Study Steps

- Revisit the recall references if any older idea still feels weak.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

# Next Study Steps

- Revisit the recall references if any older idea still feels weak.
- Rework the lecture example without looking at the notes first.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

# Next Study Steps

- Revisit the recall references if any older idea still feels weak.
- Rework the lecture example without looking at the notes first.
- Attempt the self-study practice and then answer the exit questions in full sentences.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

# Next Study Steps

- Revisit the recall references if any older idea still feels weak.
- Rework the lecture example without looking at the notes first.
- Attempt the self-study practice and then answer the exit questions in full sentences.
- Apply the idea once inside Campus Resource Manager to make the concept stick.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

# Next Lecture Preview

**Next:** Swing Basics (Components, Containers, Layouts)

- Lambdas become useful for event handlers in Swing.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).